

Name: Chandra Reddy B

## REPORT

### Sales Data Analysis – Trends, Top Performers, and Business Insights

#### Dataset Source:

Kaggle – Sales Dataset

#### Tools Used:

Python (Pandas, Matplotlib)

---

#### 1 Overview of the Dataset

The dataset used for this analysis is a raw sales dataset obtained from Kaggle. It represents retail transaction data containing order details, product information, regional sales, revenue, profit, and discounts. The dataset closely resembles real-world business data.

---

#### 2 Data Cleaning and Preparation

The following data cleaning steps were performed using Python:

- Inspected dataset shape, columns, and data types
- Handled missing values in Sales, Profit, and Discount columns
- Removed duplicate records to prevent double counting
- Converted Order Date to datetime format
- Created new derived columns:
  - **Cost = Sales – Profit**
  - **Month from Order Date**

The cleaned dataset was exported as **cleaned\_sales\_data.csv**.

---

#### 3 Analysis Performed

##### ◆ Total Sales and Profit by Month

Monthly aggregation was used to identify sales trends and seasonality.

### ◆ Top 5 Products by Sales

Products were ranked based on total sales value to identify top performers.

### ◆ Sales by Region

Regional sales aggregation highlighted geographic performance differences.

### ◆ Outlier Detection

Outliers in sales values were identified using the Interquartile Range (IQR) method.

---

## Visualizations

### Monthly Sales Trend

A line chart was created to visualize monthly sales patterns with data labels.

### Sales by Region

A bar chart was used to compare total sales across regions with data labels.

All charts were exported as high-resolution PNG files.

---

## 5 Key Business Insights

1. Sales show noticeable seasonal trends, with certain months consistently outperforming others.
  2. A small group of products contributes a significant portion of total revenue.
  3. Sales performance varies by region, indicating opportunities for targeted regional strategies.
- 

## 6 Conclusion

This analysis provides actionable insights into sales performance, customer demand, and regional contribution. The results can support better decision-making related to inventory planning, marketing focus, and regional growth strategies.